

Distance-Vector Routing

Classroom Activities

Two hands-on activities for 12 students

Activity 1: Broken Telephone Routing (~20 min)

Activity 2: Count to Infinity Speedrun (~15 min)

Materials included:

Instructor Guide • Student Handouts • Role Cards • Message Cards

INSTRUCTOR GUIDE

Activity 1: Broken Telephone Routing

Time: 20 minutes | Groups: 3 groups of 4 students

Learning Objectives

- Experience how distance-vector advertisements propagate through a network
- Practice Bellman-Ford update rule: accept if (advertised cost + link cost) < current cost
- Understand why updates from next hop must always be accepted

Setup (5 min)

1. Divide class into 3 groups of 4 students each
2. Each group arranges themselves in a LINEAR topology (see diagram)
3. Assign roles: one student is destination "A", others are R1, R2, R3
4. Give each router student a routing table sheet and message cards
5. Announce link costs (write on board or give link cost cards)

Network Topology (draw on board)

Linear topology - all link costs = 1

A — R1 — R2 — R3

Neighbors:

- A connects to R1 only (cost 1)
- R1 connects to A and R2 (cost 1 each)
- R2 connects to R1 and R3 (cost 1 each)
- R3 connects to R2 only (cost 1)

This is the SAME topology used in Activity 2, so students will be familiar with it when tracing count-to-infinity.

Running the Activity (10 min)

Round 0: Student A announces "I am destination A, cost 0" to R1 (its only neighbor)

Round 1:

- R1 updates table: Dest=A, Via=A, Cost=1
- R1 writes a message card and passes it to R2

Round 2:

- R2 receives from R1, calculates: $1 + 1 = 2$
- R2 updates table: Dest=A, Via=R1, Cost=2
- R2 writes a message card and passes it to R3

Round 3:

- R3 receives from R2, calculates: $2 + 1 = 3$
- R3 updates table: Dest=A, Via=R2, Cost=3

The network has converged! Everyone knows how to reach A.

The Twist: Link Failure! (5 min)

After convergence, announce: "The link between A and R1 just failed!"

What happens:

- R1 loses its direct path to A
- R2 still thinks A is 2 away and tells R1: "A is 2 away from me"
- R1 thinks: "Great! $2 + 1 = 3$. I can reach A through R2!"
- But wait... R2's path goes THROUGH R1! This is a loop!

Activity 2: Count to Infinity Speedrun

Time: 15 minutes | Groups: 4 groups of 3 students

Learning Objectives

- Trace through the count-to-infinity problem step by step
- Understand why it happens (routers don't know the full path)
- Learn solutions: poison reverse and max hop count (16 = infinity)

Setup

Groups of 3 at tables. Give each group the "Count to Infinity Challenge" worksheet.

The Challenge: Given the same linear topology from Activity 1, trace through exactly what happens after the link failure — this time all the way to infinity!

The Scenario (Same as Activity 1)

Linear topology: A — R1 — R2 — R3

All links cost 1. Initial state: everyone can reach A.

Initial routing tables:

R1: A via direct, cost 1

R2: A via R1, cost 2

R3: A via R2, cost 3

EVENT: Link A—R1 fails!

Part 1: The Disaster (No Fixes)

Students trace what happens without poison reverse or infinity cap:

Let students discover this problem. Don't solve it yet — this sets up Activity 2!

Discussion Questions

- Why must we always accept updates from our next-hop, even if the cost increases?
- What information does each router NOT know about the network?
- Why did R1 believe R2's advertisement even though R2's path went through R1?

6. R1 lost its route to A. R1 hears from R2: "A is 2 away." R1 thinks: "Great! $2+1=3$." R1 now believes A is 3 away via R2.
7. R1 tells R2: "A is 3 away from me." R2 thinks: "Hmm, my next-hop says 3, so $3+1=4$." R2 updates to cost 4.
8. R2 tells R3: "A is 4 away." R3 updates to 5. R2 tells R1: "A is 4 away." R1 updates to 5.
9. This continues forever! (Or until someone notices...)

Part 2: The Race

First group to correctly fill out ALL iterations until cost reaches 16 wins!

(Spoiler: it takes about 6-7 iterations for costs to reach 16)

Part 3: Apply the Fixes

Now retry with:

- **Poison Reverse:** When R2 learned about A via R1, it tells R1: "A is infinity from me"
- **Max hop count:** Any cost ≥ 16 is treated as infinity (unreachable)

Groups trace through the same scenario with fixes applied.

Discussion Questions

- Why couldn't R2 know that its path to A actually went through R1?
- What's the tradeoff of setting infinity = 16? (Answer: can't have paths longer than 15)
- Does poison reverse completely solve the problem? (Answer: no, just the 2-node case)

STUDENT HANDOUTS

Print and cut as needed

Activity 1: Router Role Cards

Cut these out and give one to each student playing a router.

ROUTER R1 Your neighbors: A, R2 Link costs: All links cost 1 Rules: • Accept if (their cost + 1) < your cost • Always accept from your current next-hop • Tell all neighbors when you update	ROUTER R2 Your neighbors: R1, R3 Link costs: All links cost 1 Rules: • Accept if (their cost + 1) < your cost • Always accept from your current next-hop • Tell all neighbors when you update
ROUTER R3 Your neighbors: R2 only Link costs: All links cost 1 Rules: • Accept if (their cost + 1) < your cost • Always accept from your current next-hop • Tell all neighbors when you update	DESTINATION A You ARE the destination! Your neighbors: R1 only Your job: • Announce "I am A, cost 0" • You don't need a routing table • Just keep announcing each round

Routing Table Worksheets

Give one to each router. Students fill in as they receive advertisements.

Router R1 Routing Table

Your neighbors: A, R2

Dest	Via	Cost

Log your updates:

Round ____ : Heard from ____ that dest ____ is cost ____.
My calculation: ____ + 1 = ____

Round ____ : Heard from ____ that dest ____ is cost ____.
My calculation: ____ + 1 = ____

Round ____ : Heard from ____ that dest ____ is cost ____.
My calculation: ____ + 1 = ____

Router R2 Routing Table

Your neighbors: R1, R3

Dest	Via	Cost

Log your updates:

Round ____ : Heard from ____ that dest ____ is cost ____ . My calculation: ____ + 1 = ____

Round ____ : Heard from ____ that dest ____ is cost ____ . My calculation: ____ + 1 = ____

Round ____ : Heard from ____ that dest ____ is cost ____ . My calculation: ____ + 1 = ____

Router R3 Routing Table

Your neighbors: R2 only

Dest	Via	Cost

Log your updates:

Round ____ : Heard from ____ that dest ____ is cost ____ . My calculation: ____ + 1 = ____

Round ____ : Heard from ____ that dest ____ is cost ____ . My calculation: ____ + 1 = ____

Round ____ : Heard from ____ that dest ____ is cost ____ . My calculation: ____ + 1 = ____

Advertisement Message Cards

Print multiple copies. Students fill these out and pass to neighbors.

ADVERTISEMENT
From Router: _____
Destination: _____
Cost: _____

ADVERTISEMENT
From Router: _____
Destination: _____
Cost: _____

ADVERTISEMENT
From Router: _____
Destination: _____
Cost: _____

ADVERTISEMENT
From Router: _____
Destination: _____
Cost: _____

ADVERTISEMENT
From Router: _____
Destination: _____
Cost: _____

ADVERTISEMENT
From Router: _____
Destination: _____
Cost: _____

COUNT TO INFINITY CHALLENGE

Activity 2 Student Worksheet

The Scenario

Linear network (same as Activity 1):

A ————— R1 ————— R2 ————— R3

All link costs = 1

Initial state (before disaster):

R1 knows: A is 1 hop away (direct link)

R2 knows: A is 2 hops away (via R1)

R3 knows: A is 3 hops away (via R2)

DISASTER: The link between A and R1 just FAILED!

Part 1: Trace the Disaster (No Fixes)

Fill in what happens at each step. First team to reach cost 16 correctly wins!

Step	R1's cost to A	R2's cost to A	R3's cost to A
Initial	1 (direct)	2 (via R1)	3 (via R2)
FAIL	??? (link down)	2	3
1			
2			
3			
4			
5			
6			
7			
8			

Hint: At each step, think about what advertisement each router sends and receives.

Part 2: Apply the Fixes!

Fix 1: Poison Reverse

Rule: When you learn a route via neighbor X, tell X that the destination is INFINITY away from you.

Why it helps: _____

Trace through the same scenario with poison reverse:

Step	R1 cost	R2 cost	R3 cost	Stable?
0				
1				
2				
3				
4				

Fix 2: Infinity = 16

Rule: Any cost ≥ 16 means "unreachable." Stop advertising.

Tradeoff: _____

Discussion Questions

1. In Part 1, what did R1 believe after hearing R2's advertisement? Why was this wrong?

2. Does poison reverse completely solve count-to-infinity? What about 3+ node loops?

3. Why is 16 specifically chosen as infinity in RIP (the real protocol)?

ANSWER KEY (Instructor Only)

Count to Infinity - Part 1 Answers

Assuming R2 advertises to R1 first each round:

Step	R1	R2	R3
Initial	1	2	3
After fail	3 (via R2)	2	3
1	3	4	3
2	5	4	5
3	5	6	5
...	...	...	...
~7	≥ 16 (inf)	≥ 16 (inf)	≥ 16 (inf)

Key insight for students:

R1 heard "A is 2 away" from R2, but R2's path to A went THROUGH R1! R2 didn't know this because distance-vector only shares (destination, cost), not the full path. This is the fundamental limitation.

Part 2 - Poison Reverse Answer

With poison reverse, R2 would tell R1: "A is infinity from me" (because R2 learned about A via R1).

So when the link fails, R1 gets no false hope from R2. R1 correctly knows A is unreachable.

R2 then learns A is unreachable from R1, tells R3, and the network stabilizes at infinity quickly.

Discussion Answers:

1. R1 believed it could reach A in 3 hops via R2. This was wrong because R2's path went through R1 itself - a routing loop.
2. Poison reverse only solves 2-node loops. With 3+ nodes (R1-R2-R3-R1), you can still get count-to-infinity because R1 might hear from R3, who heard from R2.
3. 16 is chosen as a reasonable tradeoff: small enough that count-to-infinity terminates quickly, large enough for most campus/enterprise networks. The tradeoff is RIP can't handle networks with paths longer than 15 hops.